Activity 1

Take a white sheet of paper. Construct two perpendicular lines OX, OY which meet at O, as shown in the Fig. 6.4(a).

Considering OX as X axis, OY as Y axis.

We will verify the values of $\sin\theta$ and $\cos\theta$ for certain angles θ.

Let $\theta = 30°$

Construct a line segment OA of any length such that $\angle AOX = 30°$, as shown in the Fig. 6.4(b).

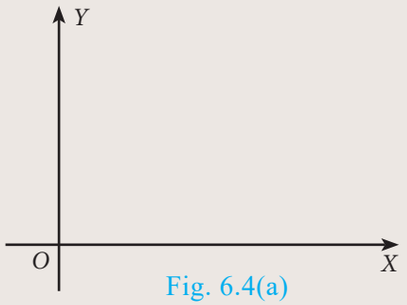


Fig. 6.4(a)

Draw a perpendicular from A to OX, meeting at B.

Now using scale, measure the lengths of AB, OB and OA.

Find the ratios $\frac{AB}{OA}, \frac{OB}{OA}$ and $\frac{AB}{OB}$.

What do you get? Can you compare these values with the trigonometric table values? What is your conclusion? Carry out the same procedure for $\theta = 45°$ and $\theta = 60°$. What are your conclusions?

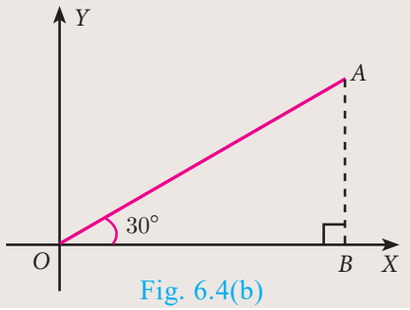


Fig. 6.4(b)

Example 6.1 Prove that $\tan^2\theta - \sin^2\theta = \tan^2\theta \sin^2\theta$

Solution $\tan^2\theta - \sin^2\theta = \tan^2\theta - \frac{\sin^2\theta}{\cos^2\theta}.\cos^2\theta$

$= \tan^2\theta(1 - \cos^2\theta) = \tan^2\theta \sin^2\theta$

Example 6.2 Prove that $\frac{\sin A}{1+\cos A} = \frac{1-\cos A}{\sin A}$

Solution $\frac{\sin A}{1+\cos A} = \frac{\sin A}{1+\cos A} \times \frac{1-\cos A}{1-\cos A}$ [multiply numerator and denominator by the conjugate of $1+\cos A$]

$= \frac{\sin A(1-\cos A)}{(1+\cos A)(1-\cos A)} = \frac{\sin A(1-\cos A)}{1-\cos^2 A}$

$= \frac{\sin A(1-\cos A)}{\sin^2 A} = \frac{1-\cos A}{\sin A}$

Example 6.3 Prove that $1 + \frac{\cot^2\theta}{1+\operatorname{cosec}\theta} = \operatorname{cosec}\theta$

Solution $1 + \frac{\cot^2\theta}{1+\operatorname{cosec}\theta} = 1 + \frac{\operatorname{cosec}^2\theta - 1}{\operatorname{cosec}\theta + 1}$ $\left[\because \operatorname{cosec}^2\theta - 1 = \cot^2\theta\right]$

$= 1 + \frac{(\operatorname{cosec}\theta + 1)(\operatorname{cosec}\theta - 1)}{\operatorname{cosec}\theta + 1}$

$= 1 + (\operatorname{cosec}\theta - 1) = \operatorname{cosec}\theta$

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